بسم الله الرحمن الرحيم

Chapter 5: Thermochemistry

thermodynamic

E ∝ chemical rxn

E

Energy

kinatic

potential

$\boxed{\frac{1}{2}mv^2}$

Electrostatic potential energy

+ −
− −
+ +

$E = \frac{KQ_1Q_2}{d}$

E : SI unit : Joul : J

$\boxed{1\ cal = 4.184\ J}$

E → w
E → q

internal energy

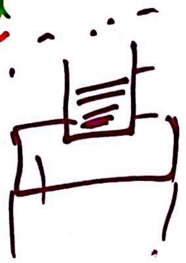
$\boxed{\Sigma(p + k)}$

state function

$\Delta E \rightarrow E_f - E_i$

System and Surrounding = universe

isolated



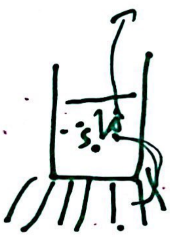
neither energy nor matter can be exchanged with surr.

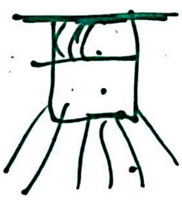
open

matter & energy can be exchanged between sys and surr.

closed

sys can exchange energy with surr but can't exchange matter





First law of thermodynamic

→ Any energy that is lost by the system must be gained by the surroundings, and vise versa.

→ Energy can be neither destroyed nor created.

→ Energy is conserved.

$$\Delta E_{universe} = \text{zero}$$

$$\Delta E_{sys} + \Delta E_{sur} = 0$$

$$\Delta E_{sys} = - \Delta E_{surr}$$

$E \rightarrow w \rightarrow$ done on the system $\rightarrow \boxed{+}$

$w \rightarrow$ done **by** the sys $\boxed{-}$

$E \rightarrow q$

$q \rightarrow$ sys gained heat $\boxed{+}$

$q \rightarrow$ sys losses heat $\boxed{-}$

$$\boxed{E = q + w}$$

$$E = q - P\Delta V$$

PV-work

$$\boxed{w = -P\Delta V}$$

by the sys	on the sys
$\Delta V = +$	$\Delta V = \boxed{-}$
w (−)	w +

$$w = -P\Delta V$$

endo بتعطيني طاقة حرارة

exo

exothermic $(-)q$

endothermic $(+q)$

Enthalpy (H) : $\boxed{\text{heat}}$ transferred between the sys and surr carried out under constant pressure.

$$\boxed{H = q_p}$$

$$\Delta E = q - P\Delta V$$

$$\Delta E = \Delta H - P\Delta V$$

$$\boxed{\Delta H = \Delta E + P\Delta V}$$

endo exo

$+$ $-$

Calculate the change in the internal energy and determine weather the Proceses is endothermic or exothermic :

$\Delta E = q + W$

$\Delta E = q - P\Delta V$

A) a system releases $\boxed{66.1 \text{ kJ}}$ of heat to its surroundings while the surroundings do 44 kJ of work on the system.

$-q$

$+$

$$\Delta E = q + w$$

$$= -66.1 + 44 \text{ F}$$

$$\boxed{\Delta E = -22.1 \text{ kJ}}$$

exothermic

b) a system absorbs 140 J of heat from the surroundings and does 85 J of work on the surrounding.

$+q$ $-W$

$\Delta E = W + q$

$= -85 + 140$

$= 55\,J$

endothemic

c) a system absorbs 7.25 kJ of heat from surrounding while it's volume remains constant.

$+q$

$\Delta E = q - \cancel{P\Delta V}$

$= 7.25\,kJ$

endothemic

Enthalpies of reactions

ΔH state func.

$\Delta H = H_{products} - H_{reactant}$

H: extensive

$\boxed{H}$

$$\left[CH_{4(g)} + 2O_{2(g)} \longrightarrow CO_{2(g)} + 2H_2O\right] \quad \Delta H = -802 \xrightarrow{\times 2} \Delta H = -1604\ kJ$$

$\times 2$

$$2CH_{4(g)} + 4O_2 \longrightarrow 2CO_{2(g)} + 4H_2O$$

$$CO_{2(g)} + 2H_2O \longrightarrow 2O_{2(g)} + CH_{4(g)} \quad \Delta H = 802\ kJ$$

$$2CO_{2(g)} + 4H_2O \longrightarrow 4O_{2(g)} + 2CH_{4(g)} \quad \Delta H = 1604\ kJ$$

Consider the following reaction:

$2Mg_{(s)} + O_{2(g)} \longrightarrow 2MgO_{(s)}$, $\boxed{\Delta H = -1204 \text{ kJ}}$

A) calculate the amount of heat transferred when 3.55g of Mg reacts at constant pressure. $MW(Mg = 24.305 \text{ g/mol})$

$\boxed{\Delta H}$

$n = \frac{\text{mass}}{\text{Molar mass}}$

$= \frac{3.55 \cancel{g} \cdot mol}{24.305 \cancel{g}}$

$= 0.146 \text{ mol}$

$2 \longrightarrow -1204$

$0.146 \longrightarrow X$

$\Delta H = -87.9 \text{ kJ}$

B) How many grams of MgO are produced during an enthulpy change of -234 kJ?

2 → -1204
X → -234

X = 0.39 mol

$n = \frac{m}{MM}$

$0.39 = \frac{m}{40.3044} \Rightarrow 15.72g$

c) How many kJ of heat are absorbed when 40.3 g of MgO is decomposed into Mg and O_2 at constant pressure?

$2MgO \rightarrow O_2 + 2Mg \quad \Delta H = 1204\ kJ$

$n = \frac{m}{MM}$

$= \frac{40.3}{40.3} = 1$

2 → 1204
1 → X

$\Delta H = 602\ kJ$

Hess Law

$$\left.\begin{array}{l} - \quad \Delta H \\ - \quad \Delta H \\ - \quad \Delta H \end{array}\right] =$$

Question 2 :

What is the value for the heat of compustion of the following rxn:

$$\cancel{CS_{2(l)}} + 3O_{2(g)} \longrightarrow CO_{2(g)} + 2SO_{2(g)} \quad \Delta H = ??$$

given :

$$C_{(s)} + O_{2(g)} \longrightarrow CO_{2(g)} \quad \Delta H_1 = -393.5\ kJ$$

$$2[S_{(s)} + O_{2(g)} \longrightarrow 1SO_{2(g)}] \quad \Delta H_2 = -296.8\ kJ$$

$$C_{(s)} + 2S_{(s)} \longrightarrow CS_{2(l)} \quad \cancel{\Delta H_3 = 87.9\ kJ}$$

$$CS_{2(l)} \longrightarrow 2\cancel{S_{(s)}} + \cancel{C_{(s)}} \quad -87.9\ kJ$$

$$\cancel{C_{(s)}} + O_{2(g)} \longrightarrow CO_{2(g)} \quad -393.5\ kJ$$

$$2\cancel{S_{(s)}} + 2O_2 \longrightarrow 2SO_{2(g)} \quad -593.6$$

$$CS_{2(l)} + 3O_2 \longrightarrow CO_{2(g)} + 2SO_{2(g)}$$

$$\Delta H = -1075\ kJ$$

Enthalpies of Formation

$\boxed{\Delta H_F}$ - Enthalpies of Formation.
- heat of formation.

"Enthalpy change associated with the formation of a compound from it's constituent elements."

$\boxed{\Delta H^\circ_{rxn}}$ - standard enthalpy change of rxn

"the enthalpy change ~~associated with the formation of a compound~~ when all the reactants and products in their standard state."

$\boxed{\Delta H^\circ_F}$ standard enthalpy of formation

"change in enthalpy for the rxn that forms 1 mole of the compound from it's elements in their most stable form at standard condition."

√ $C(graphite) + 3H_{2(g)} + \frac{1}{2} O_{2(g)} \rightarrow \boxed{1} C_2H_5OH$

$C(graphite) + 6H_{2(g)} + O_{2(g)} \rightarrow 2 C_2H_5OH$ ✗

Standard conditions :
- 1 atm.
- 25 C°

$Br_{2(L)}$ / $O_{2(g)}$ / $Fe_{(s)}$

$N_{2(g)}$ / C graphite

$$\Delta H^\circ_{rxn} = \sum n \Delta H^\circ_f (\text{Products}) - \sum m \Delta H^\circ_f (\text{reactants})$$

Ex : Calculate heat of rxn for the combustion of propane gas giving carbon dioxide and water. given :

$$1\,C_3H_{8\,(g)} + 5\,O_{2\,(g)} \longrightarrow 3\,CO_{2\,(g)} + 4\,H_2O_{(l)}$$

ΔH°_f -103.85 -393.5 -285.8

$$\Delta H^\circ_{rxn} = \left[(3 \times -393.5) + (4 \times -285.8)\right] - \left[-103.85 + 5 \times 0\right]$$

$$= -2219.85 \text{ KJ}$$

Complete combustion of 1 mole of C_3H_6O liberates 1790 KJ. Given that the $\Delta H_f^\circ[CO_2] = -393.5$ and $\Delta H_f^\circ[H_2O] = -285.8$ KJ/mol.
Calculate $\Delta H_f^\circ(C_3H_6O_2)$.

$$4O_2 + C_3H_6O \longrightarrow 3CO_2 + 3H_2O \qquad \Delta H = -1790 \text{ kJ}$$

$$\begin{array}{cccc} 0 & & -393.5 & -285.8 \end{array}$$

$$-1790 = [(3 * -393.5) + (3 * -285.8)] - X$$

~~1466.9 KJ~~

-247.9 KJ

Heat Capacity (C)

the amount of heat required to raise the temp of substance by 1K or 1°C.

$$\text{J/K or J/C} \leftarrow C = \frac{q \leftarrow \text{J}}{\Delta T \leftarrow \text{K or °C}}$$

$$q = C \cdot \Delta T$$

Specific heat (Cs)

the amount of heat required to raise the temp of 1g of a substance by on degree temp.

$$\text{J/g·K, J/g·°C} \leftarrow C_s = \frac{q \leftarrow \text{J}}{\text{g} \rightarrow m \cdot \Delta T \leftarrow \text{K or °C}}$$

$$q = \underbrace{C_s \cdot m}_{} \cdot \Delta T$$
$$= C \cdot \Delta T$$

$$C_s = \frac{1}{m} \cdot C$$

Molar heat Capacity (Cm)

the amount of heat required to raise the temp of 1 mole of substance by 1K° or 1°C.

$$\text{J/K·mol, J/K°·mol} \leftarrow C_m = \frac{q \leftarrow \text{J}}{\text{mol} \rightarrow n \cdot \Delta T \leftarrow \text{°C or K}}$$

$$q = C_m \cdot n \cdot \Delta T$$
$$= C_s \cdot \text{Molar mass}$$

$$n = \frac{\text{mass}}{MM}$$

$$C_m = \frac{1}{n} C$$
$$= \frac{MM}{m} C$$

$$\boxed{C_m = C_s \cdot MM}$$

~~$n = \frac{\text{mass}}{\text{molar mass}}$~~

~~$C_m = \frac{m}{MM} \cdot \Delta T$~~

~~$C_m = \frac{1}{n} C$~~

~~$C_m = \frac{MM}{m} \cdot C$~~

Calorimetry : measurment of heat flow → $\boxed{\Delta H}$

$q_{sys} = -q_{surr}$

Calorimeter : device used to measure heat flow

$(C_s \cdot m \cdot \Delta T)_{sys} = -(C_s \cdot m \cdot \Delta T)_{surr}$

$\Delta H = \dfrac{q}{\Delta n}$

Coffee-cup

- constant P. ($q_p = \Delta H$)
- rxns: Acid-base, redox, precipitation

$$* \boxed{q_{rxn} = -q_{solution} = -C_s \cdot \text{total mass} * \Delta T}$$

C_s: 4.18 ; ΔT: ??

total mass = total volume × d = 1 g/mL

$C_s = 4.18$ J/mol.K

$d = 1$ g/mL

Bomb

- constant V
  $[q_v = \Delta E = \Delta H]$
- combustion rxn

$$* \boxed{q_{rxn} = -C_{cal} * \Delta T}$$

Q: IF 209 J is required to inc the temp of 50g of water by 1k, calculate the specific heat of water.

$$C_s = \frac{q}{m \cdot \Delta T} = \frac{209}{50 * 1} = 4.18 \ J/g.k$$

Q: IF 24.2 KJ is used to warm a piece of Al. with a mass of 250g, what is the final temp of the Al if the intial temp is 5°C? $\boxed{C_s = 0.9 \ J/g.k}$

$$C_s = \frac{q}{m * \Delta T}$$

$$0.9 = \frac{24.2 * 10^3}{250 \cdot \Delta T} \rightarrow \Delta T = \cancel{107} \ 108$$

$$\Delta T = T_f - T_i$$

$$108 = T_f - 5$$

$$\boxed{113 \ C^\circ = T_f}$$

Specific heat of some Substance at 298 K :

Substance	Cs
$N_{2(g)}$	1.04
$Al_{(s)}$	0.902
$Fe_{(s)}$	0.45
$Hg_{(l)}$	0.14
$H_2O_{(l)}$	4.184
$H_2O_{(s)}$	2.06
$CH_{4(g)}$	2.2
$CO_{2(g)}$	0.84

A) which one requires the smallest amount of E to inc the temp of 50 g by 10 K ?

$\downarrow Cs = \frac{q\downarrow}{m * \Delta T}$

$Hg_{(l)}$

B) What is the molar heat capacity of CO_2 ?

MW = 44

$Cm = \frac{q}{n * \Delta T}$

$= Cs \cdot MM = 0.84 * 44 = 36.96$ J/K mol

c) how many kJ of heat required to raise temp of 10 Kg of Al from 24.6 K to 46.2 K ?

10^4

$\Delta T = 21.6$

$Cs = \frac{q}{m * \Delta T}$

$q = 0.902 * 10^4 * 21.6 = 194.8$ kJ

IF 30 g piece of copper pipe at 80 C is placed in 100 g of water at 27 C°

What is the final temp? [C_s of water = 4.18 J/g.k
C_s of copper = 0.385 J/g.k]

$$q_{sys} = -q_{surr}$$

$$(C_s \cdot m \cdot \Delta T)_{sys} = (C_s \cdot m \cdot \Delta T)_{surr}$$

$$0.385 \times 30 \times (T_F - 80) = 4.18 \times 100 \times (T_F - 27)$$

X X

$$\boxed{T_F = 25.5}$$

50 ml of 1 M HCl at 25 °C were mixed with 50 ml of 1M NaOH also at 25 °C in styrofoam cup calorimeter. After the mixing process, the thermometer reading was 31.9 °C. [d = 1 g/ml C_s = 4.18 J/K.g]

Calculate:

A the energy involved in the rxn.

$d = \frac{m}{V}$

$q_{rxn} = -C.S * \text{total mass} * \Delta T$

$= 4.18 \times 100\,g * (31.9 - 25)$

$= -2.9\ KJ$

$1 = \frac{m}{100}$

$m = 100\,g$

total mass = total volume × d.

B the enthalpy per moles of hydrogen ions used.

$\Delta H = \frac{q}{n}$

$= \frac{-2.9}{50 * 10^{-3}} = -58\ KJ/mol$

$M = \frac{n}{V(L)}$

$\rightarrow n = 1 * 50 * 10^{-3}$

In an experiment 1.5 g of $Mg_{(s)}$ was combined with 125 mL of 1M HCl.
The initial tempreture was 25 °C and the final temp was 72.3 °C. Calculate:
$MM(Mg) = 24.305\ g/mol$.

A the heat involved in the rxn.

$$q = -C_s * \text{total mass} * \Delta T$$

$$= -4.18 * [1.5 + 125*1] * [72.3 - 25]$$

$$= -25\ kJ$$

B the enthalpy of rxn in terms of the number of moles of $Mg_{(s)}$ used.

$$n = \frac{mass}{M.M}$$

$$= \frac{1.5}{24.305} = 0.06\ mol$$

$$\Delta H = \frac{q}{n}$$

$$= \frac{-25}{0.06}$$

$$\boxed{\Delta H = -405\ kJ/mol}$$

When 5.03 g of solid potassium hydroxide (MM = 56.1056 g/mol) are dissolved in 100 ml of D.W in coffee-cup calorimeter, the temp of liquid inc from 23 °C to 34.7 °C. The density of water in this temp range is 0.9969 g/mL. What is ΔH solu? (C_s = 4.18 J/g.k).

$$q_{sol} = C_s * \text{total mass} * \Delta T$$

$$= 4.18 * [5.03 + (100 * 0.9969)] * (34.7 - 23)$$

$$= 5.12 \text{ kJ}$$

$$\Delta H = \frac{q}{n}$$

$$= \frac{5.12}{\left(\frac{5.03}{56.1056}\right)} = \boxed{57.1 \text{ kJ/mol}}$$

The combustion of $\overset{m_1}{0.579\,g}$ of benzoic acid in bomb calorimeter caused a $\overset{\Delta T_1}{2.08}$ °C inc in the temp of calorimeter. The chamber was then emptied and recharged with $\underset{m_2}{1.732\,g}$ of Glucose and excess oxygen. Ignition of the glucose resulted in a temp inc of $\overset{\Delta T_2}{3.64}$ °C. What ΔH_{comb} of glucose?

[ΔH_{com} of B.A = -26.38 kJ/g , $MM(O_2) = 16$ g/mol , $MM(BA) = 122.12$ g/mol

$MM(Glucose) = 180.156$ g/mol]

$\boxed{q = -C_{cal} * \Delta T}$ $\boxed{B.A}$

$$\Delta H = \frac{q}{n} \rightarrow q = -26.38 * \frac{0.579}{122.12} = -0.13\ kJ$$

$$q = -\boxed{C_{cal}} * \Delta T$$

$$-0.13 = -C_{cal} * 2.08$$

$$\boxed{C_{cal} = 0.06}$$

$\boxed{G}$

$$q = -C_{cal} * \Delta T$$

$$= -0.06 * 3.64 = -0.2184\ kJ$$

$$\therefore \Delta H = \frac{q}{n} = \frac{-0.2184}{\left(\frac{1.732}{180.156}\right)} = \boxed{2.78 * 10^4\ J/mol}$$

Chapter 5 ✓

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